Tejan Karmali

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Research Interests: Generative Models, Reinforcement Learning

I am interested in researching on generative models that synthesize training data for reinforcement learning. I am **seeking a Summer 2027 research internship**.

Education

Stanford University, PhD in Computer Science (Advisor: Prof. Kayvon Fatahalian) Sept 2024 - Present

Indian Institute of Science, M.Tech. (Research) in Computational and Data Sciences 2019 - 2022

Research Experience

Research Intern - Applied Intuition Summer 2026
- Developed a Flow-matching based Generative Perception Error Model, enabling self-play policy learning and inference in perception space.

Graduate Researcher - Stanford University Sept 2024 - Present
- Built a real-time game solver using a hybrid PPO+GoExplore algorithm with 8K parallel GPU simulations, answering constrained solvability queries for 3D obstacle courses in seconds (**SIGGRAPH Asia 2026**).
- Applied Maximum Entropy RL to generate diverse, multi-modal solution sets.

Research Engineer II - Avataar Oct 2023 - July 2024
- Developed a cascaded LRM+Zero123 pipeline for novel view synthesis at 512 resolution.
- Built a data curation pipeline over ∼800K 3D assets using CLIP embeddings across 10M rendered images, enabling labellers to curate a 240K high-quality subset of Objaverse in 10 minutes and improving downstream NVS quality by 25% on LPIPS.

Pre-Doctoral Researcher - Google Research Apr 2022 - May 2023
- Developed a transformer-based semantic segmentation model for high-resolution mapping of agricultural fields from multitemporal, low-resolution satellite imagery.

Publications

- *Providing Rapid Design Feedback for 3D Obstacle Course Games Using Constrained Solvability Queries*
  Majercik Z., Zhang S., Wang W., **Karmali T.**, Zhou F., Yuan Y., Chou J., Agrawala M., Fatahalian K. **SIGGRAPH Asia 2026**
- *Hierarchical Semantic Regularization of Latent Spaces in StyleGANs*
  Karmali T., Parihar R., Agrawal S., Rangwani H., Jampani V., Singh M., Babu R.V. **ECCV 2022**
- *LEAD: Self-Supervised Landmark Estimation by Aligning Distributions of Feature Similarity*
  Karmali T.*, Atrishi A.*, Harsha S.S., Agrawal S., Jampani V., Babu R.V. **WACV 2022**
- *NoisyTwins: Class-Consistent and Diverse Image Generation Through StyleGANs*
  Rangwani H.*, Bansal L.*, Sharma K., **Karmali T.**, Jampani V., Babu R.V. **CVPR 2023**
- *PreciseControl: Enhancing Text-to-Image Diffusion Models with Fine-Grained Attribute Control*
  Parihar R., Sachidanand V.S., Mani S., **Karmali T.**, Babu R.V. **ECCV 2024**
- *Improving GANs for Long-Tailed Data Through Group Spectral Regularization*
  Rangwani H., Jaswani N., **Karmali T.**, Jampani V., Babu R.V. **ECCV 2022**
- *Deep Implicit Surface Point Prediction Networks*
  Venkatesh R., **Karmali T.**, Sharma S., Ghosh A., Babu R.V., Jeni L., Singh M. **ICCV 2021**
- *Everything is There in Latent Space*
  Parihar R., Dhiman A., **Karmali T.**, Babu R.V. **ACM MM 2022**

Skills

Frameworks & Languages: PyTorch, JAX, Python, CUDA
Reinforcement Learning: PPO, Go-Explore, MaxEnt RL, large-scale GPU-parallel simulation
Generative Models: Flow Matching, Diffusion Models, GANs, Latent Space Methods, Novel View Synthesis
3D Vision & Rendering: NeRF, 3D Gaussian Splatting, Implicit Neural Representations, Differentiable Rendering

Awards & Service

Facebook AI Research International Scholarship for **DPhil** at **Oxford University** (declined) 2023
Outstanding Reviewer at **CVPR** 2022
2x **Google Summer of Code** recipient 2018-2019
Teaching Assistant, **CS248A** Computer Graphics, Stanford 2026
Reviewer: CVPR, ICLR, NeurIPS 2022-2024